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Title: Carbon-Aware AI Scheduling

Abstract

As large-scale machine learning training and workloads accelerate globally, the carbon footprint of AI has emerged as a major environmental hazard. Right now, AI executes tasks immediately upon submission, not even taking carbon intensity as a factor. This paper introduces a Python and pandas scheduling system that shifts flexible AI workloads such as LLM fine-tuning, vector embedding generation, and data ingestion and aligns with regional renewable energy peaks and Time-of-Use electricity tariffs. The scheduler was tested across a 30-day historical data of the British National Grid (mean intensity: 107.82 gCO2/kWh, Coefficient of Variation: 44.8%). Our findings show that carbon-aware orchestration reduces carbon emissions by 2,647.51 grams per 145 different tasks while staying within the task deadline. Furthermore, results highlight that high-power, long-duration tasks like 7-billion-parameter LLM fine-tuning serve as the primary source of the carbon emissions savings by the scheduler, with an average reduction of 78.86 grams of CO2 per task

1. Introduction & Reasons

1.1 The Environmental Effect of AI:

Modern-day LLMs require a large amount of processing power, resulting in continuous, high-scale power draws across multiple data centers. While much of the green AI research focuses on algorithmic efficiency (such as model quantization and pruning), infrastructure-level management is widely ignored. Electrical grids experience moderate carbon intensity fluctuations due to wind and solar generation. During the evening, grids mostly rely on fossil-fuel plants, which causes carbon emissions per kilowatt-hour to spike significantly.

1.3 Research Contributions

This paper investigates whether flexible AI tasks can be delayed to use renewable energy. The primary contributions of this work are:

A Multi-factor Scheduling Framework: A Python-based decision engine that balances hardware power, task deadlines, carbon intensity, and ToU electricity tariffs. **Grid Analysis:** A 30-day evaluation of the British National Grid establishing carbon volatility metrics, confirming a Variation of 44.8% over 1,440 half-hourly records. **Workload-Specific Benchmarking:** An assessment across 5 distinct AI tasks, proving that high-energy, flexible jobs achieve more carbon savings compared to lightweight AI tasks.

2. Methodology & Program Design:

2.1 Grid Intensity Modeling & Data Ingestion

To model realistic electrical grid carbon intensity, the scheduling framework processes historical data from the British National Grid. Due to renewable energy generation (wind and solar), grid carbon intensity exhibits high temporal variance. To capture this behavior over a statistically significant timeframe, the simulation utilizes 30 days of half-hourly records (1,440 data points), establishing a mean intensity of $107.82\ gCO_2/kWh$ and a Standard Deviation of 48.31, yielding a Coefficient of Variation (CV) of 44.8%

2.2 Workload data & Shifting Logic

The scheduler models 5 distinct AI hardware execution states, varying across power draw (kW), estimated runtime, and maximum allowable delay (deadline windows):

1. **Data Scraper** ($0.055kW$, $0.75hrs$, $12\ -\ hour$ deadline)
2. **Vector Embedding Generation** ($0.120kW$, $0.50hrs$, $6\ -\ hour$ deadline)
3. **7B LLM Fine-Tuning / LoRA** ($0.350kW$, $3.00hrs$, $18\ -\ hour$ deadline)
4. **Batch Inference** ($0.250kW$, $1.00hr$, $2\ -\ hour$ deadline)
5. **FAISS Vector Store Indexing** ($0.090kW$, $0.25hrs$, $8\ -\ hour$ deadline)

Upon task submission at timestamp t, the scheduler evaluates carbon intensity values across the forward-looking interval $[t, t\ +\ deadline]$. Utilizing Pandas dataframes, the algorithm identifies minimum intensity timestamp ($t_{optimal}$) within the permissible window and shifts execution to that time frame, ensuring zero deadline violations.

3.2 Task-Specific Efficiency Breakdown

Analysis of individual task performance indicates that savings scale exponentially with hardware power draw and task duration:

Task Name	Hardware Profile	Avg CO2 Saved per Run	30-Day Contribution Profile
7B LLM Fine-Tuning	GPU ($0.350kW, 3.0hrs$)	$78.86g$	Primary Ecological Driver
Batch Inference	GPU ($0.250kW, 1.0hr$)	$5.49g$	Moderate Impact

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Vector Embedding Generation	GPU ($0.120kW, 0.5hrs$)	$3.23g$	Moderate Impact
Data Scraper	CPU ($0.055kW, 0.75hrs$)	$2.42g$	Baseline Contribution
FAISS Vector Store Indexing	CPU/RAM ($0.090kW, 0.25hrs$)	$1.30g$	Minimal Impact

As visualized in Figure 1 (assets/carbon_curve.png), high-wattage GPU training workloads leverage flexible 18-hour windows to bypass evening fossil-fuel peaks, capturing the highest carbon reductions.

4. Discussion & Limitations

4.1 Trade-Offs Between Delay and Carbon Reduction

The experimental findings confirm that carbon-aware task scheduling successfully schedules non-urgent workloads from carbon-intensive periods. However, this optimization introduces a deliberate trade-off: **computational latency in exchange for ecological benefit**. As demonstrated by short-deadline tasks (e.g., *Batch Inference* with a 2-hour window), tightly constrained tasks severely limit the scheduler's ability to search for renewable energy sections, resulting in lower average savings ($5.49g$). Conversely, highly flexible workloads (e.g., *7B LLM Fine-Tuning* with an 18-hour window) grant the scheduler the temporal freedom necessary to bypass peak fossil-fuel generation intervals entirely, yielding an average reduction of $78.86g$ per run.

4.2 Limitations & Real-World Deployments

While this simulation models realistic grid variability (CV = 44.8%) and Time-of-Use tariffs, several real-world engineering constraints must be addressed in future iterations:

- **Hardware State Transitions:** The model assumes instantaneous power scaling upon task activation, ignoring minor thermal ramp-up.
- **Geographic Heterogeneity:** The current framework is optimized for the British National Grid; the results of the scheduler may vary from area to area.

5. Conclusion & Future Work

5.1 Conclusion

As artificial intelligence infrastructure scales globally, sustainable software engineering must move beyond purely algorithmic optimization and move to infrastructure-level flexibility and renewable sources. This research demonstrated that a lightweight, multi-objective scheduling engine can dynamically schedule AI workloads against British National Grid, achieving a cumulative savings of $2,647.51 grams of CO_2$ and **$2.85 USD** across 145 simulated task executions over 30 days. By proving that heavy GPU training workloads act as the primary drivers of carbon reduction, this work establishes a scalable framework for green AI deployment.

5.2 Future Work

Future extensions of this project will explore:

1.Multi-Region Cloud Federation: Expanding the scheduler to dynamically route workloads across data centers in different geographic zones (e.g., shifting compute from high-carbon fossil regions to wind-surplus regions).

2.Predictive Grid Forecasting: Integrating an AI time-series model to predict carbon intensity spikes 24 to 48 hours ahead, allowing the system to plan optimal execution windows proactively.

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